

Carlos De Manuel Vicente

Data & Analytics Engineer

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Data & Analytics Engineer at a global automotive manufacturer. I own the path from raw operational data to the products people use: Python pipelines orchestrated with Prefect, tested Snowflake and dbt models, Power BI and Streamlit on top, running on AWS. Used across 20+ plants in North America, APAC and EMEA, from top management to the shop floor.

EXPERIENCE

Data & Analytics Engineer Jan 2025 – Present
RESRG Automotive · global automotive manufacturer Liria, Spain (hybrid)

- Replaced plant-level Excel trackers with a governed Snowflake and dbt layer, now the single source for **20+ plants** in three regions: **130+ KPIs** and their targets defined once, in a Streamlit app with dbt tests on every write-back.
- Built and orchestrated Python ETL with Prefect from REST APIs, email and operational systems, including near-real-time inventory data; made stock loading **~90% faster** at **~30% lower cost**, mostly by cutting idle time.
- Delivered the Power BI dashboards and Streamlit apps that plants and leadership run on: direct labour efficiency, scrap, inventory days on hand, material consumption and an energy forecast for purchasing, on Snowflake models I designed and tuned.
- Deployed the apps to production on AWS (EC2, S3, EFS) with Docker, CI/CD and Prometheus monitoring.

Data Scientist (internship) Nov 2023 – Jul 2024
Biomedical Data Science Lab, ITACA, UPV Valencia, Spain

- Built a Python framework to audit healthcare ML models against **Trustworthy AI** principles: fairness, robustness and explainability (SHAP), the basis of my BSc thesis.

PROJECTS

Job Market Intelligence · [case study](#) · [github.com/carlosdmv7/job-market-intelligence](#)
Reads EU data-job ads every day and ranks them against a CV: six free APIs, LLM enrichment that went from **23 to 180** ads a day on the same free quota, dbt on MotherDuck, Prefect Cloud, GitHub Actions CI and Streamlit.

Spanish Housing Radar · [case study](#) · [github.com/carlosdmv7/spanish-housing-radar](#)
Finds under-priced flats in València by scoring every listing against its neighbourhood: weekly scraping, dbt with **162 data tests**, Prefect Cloud, Docker, GitHub Actions CI and Streamlit.

SKILLS

Data modelling & warehousing	Snowflake, dbt, DuckDB, dimensional modelling (Kimball)
Languages & APIs	Python, SQL, Bash, Jinja, DAX, REST APIs
Platform & CI/CD	AWS, Prefect, Airflow, GitHub Actions, Git, Azure DevOps, Docker, Make
Quality & monitoring	pytest, Pydantic, ruff, mypy, Prometheus
BI & apps	Power BI, Streamlit, Microsoft Fabric
AI & ML	LLM APIs (Gemini, Azure OpenAI), LLM evaluation, scikit-learn, SHAP

EDUCATION

MSc, Data Analysis, Process Improvement & Decision Support Engineering · UPV, Valencia
BSc, Biomedical Engineering (ICT specialisation) · UPV, Valencia
Thesis: a Trustworthy AI framework for healthcare, graded 10/10 · top 25% of the year

LANGUAGES

Spanish (native) · English (C1, Cambridge, 194)